

Model Analysis Report

Flood Polynomial Model

MKM-FPO-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: $P(\text{flood})$ sensitivity to water level and storm hour (severe threshold $s = 6.58$ m AOD).

Water Level (m)	H0	H12	H24	H48	H96	H167
4.00	0.00	0.00	0.00	0.00	0.00	0.00
4.50	0.00	0.00	0.00	0.00	0.01	0.02
5.00	0.00	0.00	0.01	0.02	0.07	0.15
5.50	0.01	0.02	0.05	0.13	0.33	0.53
6.00	0.07	0.14	0.25	0.45	0.71	0.85
6.30	0.20	0.34	0.49	0.69	0.87	0.94
6.58	0.42	0.58	0.71	0.84	0.94	0.97
7.00	0.72	0.82	0.89	0.95	0.98	0.99
7.50	0.93	0.96	0.97	0.99	1.00	1.00

Interpretation:

[nosep]At sub-alert levels (4.0–5.0 m), $P(\text{flood}) \approx 0$ regardless of hour, consistent with the GBM surface. At the severe threshold (6.58 m), $P(\text{flood})$ ranges from 0.42 at hour 0 (high uncertainty, storm just started) to 0.97 at hour 167 (near certainty, storm almost over). Above the severe level (7.0+ m), $P(\text{flood})$ saturates toward 1.0, reflecting overwhelming evidence of flood breach. The hour dimension provides meaningful discrimination primarily in the transition zone (5.5–6.6 m), where early-hour uncertainty is largest.

Table 2: Polynomial vs GBM comparison at selected grid points (GAUGE-dc5e88fb, $\Delta w = 0$).

Level (m)	Hour	GBM	Poly	$ \Delta $
5.00	24	0.01	0.01	0.00
5.50	48	0.14	0.13	0.01
6.00	24	0.27	0.25	0.02
6.30	48	0.72	0.69	0.03
6.58	96	0.93	0.94	0.01
7.00	48	0.96	0.95	0.01

The absolute difference between the polynomial and GBM predictions is consistently below 0.05 across the surface, with the largest residuals concentrated in the transition zone around the alert threshold.

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 3: Stress testing scenarios for Flood Polynomial Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 4: Performance metrics for Flood Polynomial Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.